

■ 시나리오 별 백업 복구(User Managed)**KEY WORDS**

- <시나리오 12> Backup 에 Redo log files 이 존재하지 않는 경우
 - <시나리오 13> datafile, redo log file 은 손상되지 않고 control file 만 손상되었을 경우 복구(Binary)
 - <시나리오 14> datafile, redo log file 은 손상되지 않고 control file 만 손상되었을 경우 복구(Trace)
 - <시나리오 15> datafile, redo log file 은 손상되지 않고 control file 만 손상되었을 경우 복구(정상종료)
 - <시나리오 16> datafile, redo log file 은 손상되지 않고 control file 만 손상되었을 경우 복구(비정상종료)
 - <시나리오 17> datafile, control file 손상되었을 경우 복구(데이터베이스 정상 종료, 리두정보 O)
-

<<시나리오 12>> Backup 에 Redo log files 이 존재하지 않는 경우

-- 복구 정보(체크포인트) 확인

- 백업 진행

```
SYS@orcl> shutdown immediate
```

```
SYS@orcl> ! mkdir -p backup/noredo
```

```
SYS@orcl> !cp -v /u01/app/oracle/oradata/ORCL/*.{dbf,ctl} /home/oracle/backup/noredo/  
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> '/home/oracle/backup/noredo/sysaux01.dbf'  
'/u01/app/oracle/oradata/ORCL/system01.dbf' -> '/home/oracle/backup/noredo/system01.dbf'  
'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> '/home/oracle/backup/noredo/temp01.dbf'  
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> '/home/oracle/backup/noredo/undotbs01.dbf'  
'/u01/app/oracle/oradata/ORCL/users01.dbf' -> '/home/oracle/backup/noredo/users01.dbf'  
'/u01/app/oracle/oradata/ORCL/control01.ctl' -> '/home/oracle/backup/noredo/control01.ctl'  
'/u01/app/oracle/oradata/ORCL/control02.ctl' -> '/home/oracle/backup/noredo/control02.ctl'
```

```
SYS@orcl> startup
```

```
SYS@orcl> create table hr.emp1 tablespace users  
2 as select * from hr.employees;
```

Table created.

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> /
```

System altered.

SYS@orcl> /

SYS@orcl> shut immediate

- 장애 발생

SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/users01.dbf

-rw-r-----. 1 oracle dba 5251072 May 6 15:07 /u01/app/oracle/oradata/ORCL/users01.dbf

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf

SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/users01.dbf

ls: cannot access /u01/app/oracle/oradata/ORCL/users01.dbf: No such file or directory

SYS@orcl> startup

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 7 - see DBWR trace file

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

SYS@orcl> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME
CON_ID				

7	ONLINE	ONLINE	FILE NOT FOUND	0

SYS@orcl> select file#, name, checkpoint_change#, status from v\$datafile;

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS
1	/u01/app/oracle/oradata/ORCL/system01.dbf	3268412	SYSTEM
2	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	3268412	ONLINE
3	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	3268412	ONLINE
7	/u01/app/oracle/oradata/ORCL/users01.dbf	3268412	ONLINE

SYS@orcl> alter database datafile 7 offline drop;

Database altered.

```
SYS@orcl> select file#, name, checkpoint_change#, status from v$datafile;
```

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS
1	/u01/app/oracle/oradata/ORCL/system01.dbf	3268412	SYSTEM
2	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	3268412	ONLINE
3	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	3268412	ONLINE
7	/u01/app/oracle/oradata/ORCL/users01.dbf	3268412	OFFLINE

```
SYS@orcl> alter database open;
```

Database altered.

문제가 발생한 데이터파일의 최근 백업본을 찾아 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/noarch/users01.dbf /u01/app/oracle/oradata/ORCL/
```

백업 이후 변경정보(리두) 적용

```
SYS@orcl> recover tablespace users;
```

ORA-00279: change 2306033 generated at 05/05/2025 18:10:59 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_100_1199933387.dbf

ORA-00280: change 2306033 for thread 1 is in sequence #100

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

cancel

Media recovery cancelled.

-- log switch 로 인한 정보 없음으로 오류 발생

```
SYS@orcl> alter database datafile 7 online;
```

```
alter database datafile 7 online
```

*

ERROR at line 1:

ORA-01113: file 7 needs media recovery

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

- 완전 복구 시도 했지만 redo 정보가 없어서 실패 발생

- 완전 복구 실패시 무조건 불완전 복구 방식을 시도.

```
SYS@orcl> shutdown abort
```

ORACLE instance shut down.

최근 백업본을 restore (모든 데이터파일, 컨트롤 파일)

```
SYS@orcl> !cp /home/oracle/backup/noredo/*.{ctl,dbf} /u01/app/oracle/oradata/ORCL/
```

- startup mount

```
SYS@orcl> startup mount
```

- 백업 컨트롤 파일을 이용한 cancel base recovery 시도

```
SYS@orcl> recover database until cancel using backup controlfile;
```

```
ORA-00279: change 2310249 generated at 05/06/2025 14:01:30 needed for thread 1
```

```
ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_100_1199933387.dbf
```

```
ORA-00280: change 2310249 for thread 1 is in sequence #100
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
CANCEL
```

```
Media recovery cancelled.
```

```
SYS@orcl> alter database open;
```

```
alter database open
```

```
*
```

```
ERROR at line 1:
```

```
ORA-01589: must use RESETLOGS or NORESETLOGS option for database open
```

-- 그냥 오픈하면 오류 발생

```
SYS@orcl> alter database open resetlogs;
```

```
Database altered.
```

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
OPEN
```

```
SYS@orcl> select * from v$log;
```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS	FIRST_CHANGE#
FIRST_TIME	NEXT_CHANGE#							

NEXT_TIME	CON_ID							

1	1	1	10485760	512	1	NO	CURRENT	2310250 07-
MAY-25	1.8447E+19							
	0							

```

2          1          0  10485760      512      1 YES UNUSED      0
          0
                0

3          1          0  10485760      512      1 YES UNUSED      0
          0
                0

```

-- sequence#0이 1 부터 다시 시작한다.

* 반드시 백업받기

```
SYS@orcl> alter system switch logfile;
```

```
SYS@orcl> /
```

```
SYS@orcl> /
```

```
SYS@orcl> /
```

```
SYS@orcl> select * from v$log;
```

```

SYS@orcl> !cp -v /u01/app/oracle/oradata/ORCL/*.ctl,dbf,log /home/oracle/backup/noarch/
'/u01/app/oracle/oradata/ORCL/control01.ctl' -> '/home/oracle/backup/noarch/control01.ctl'
'/u01/app/oracle/oradata/ORCL/control02.ctl' -> '/home/oracle/backup/noarch/control02.ctl'
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> '/home/oracle/backup/noarch/sysaux01.dbf'
'/u01/app/oracle/oradata/ORCL/system01.dbf' -> '/home/oracle/backup/noarch/system01.dbf'
'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> '/home/oracle/backup/noarch/temp01.dbf'
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> '/home/oracle/backup/noarch/undotbs01.dbf'
'/u01/app/oracle/oradata/ORCL/users01.dbf' -> '/home/oracle/backup/noarch/users01.dbf'
'/u01/app/oracle/oradata/ORCL/redo01.log' -> '/home/oracle/backup/noarch/redo01.log'
'/u01/app/oracle/oradata/ORCL/redo02.log' -> '/home/oracle/backup/noarch/redo02.log'
'/u01/app/oracle/oradata/ORCL/redo03.log' -> '/home/oracle/backup/noarch/redo03.log'

```

resetlogs open 시점

- 불완전한 미디어 복구(incomplete media recovery)
- 백업받은 컨트롤 파일을 이용한 복구
- 컨트롤파일을 새로 생성한 복구

<<시나리오 13>> datafile, redo log file 은 손상되지 않고 control file 만 손상되었을 경우 복구(Binary)

SYS@orcl> startup

SYS@orcl> create table hr.emp3 tablespace users as select * from hr.employees;

Table created.

SYS@orcl> alter system switch logfile;

System altered.

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
	1	7	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
NO	2324655	2324658			
	2	8	/u01/app/oracle/oradata/ORCL/redo02.log	10	ACTIVE
NO	2324658	2327378			
	3	9	/u01/app/oracle/oradata/ORCL/redo03.log	10	CURRENT
NO	2327378	1.8447E+19			

■ 장애 발생

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl

SYS@orcl> startup

ORA-01081: cannot start already-running ORACLE - shut it down first

SYS@orcl> shut abort

ORACLE instance shut down.

SYS@orcl> startup

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

ORA-00205: error in identifying control file, check alert log for more info

SYS@orcl> select status from v\$instance;

STATUS

STARTED

SYS@orcl> show parameter control_files

NAME	TYPE	VALUE
control_files	string	/u01/app/oracle/oradata/ORCL/ /control01.ctl

SYS@orcl> shutdown abort

ORACLE instance shut down.

■ 최근 백업 컨트롤 파일 찾아 restore

SYS@orcl> ! cp -v /home/oracle/backup/noarch/control01.ctl /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'

SYS@orcl> startup mount

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

- 과거 백업 컨트롤 파일 이용해서 복구작업 수행(리두 O)

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
	1	7	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
NO	2324655	2324658			
	2	8	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
NO	2324658	1.8447E+19			
	3	6	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
NO	2324652	2324655			

SYS@orcl> recover database using backup controlfile;

ORA-00279: change 2325498 generated at 05/07/2025 23:17:09 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_8_1200525254.dbf

ORA-00280: change 2325498 for thread 1 is in sequence #8

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

AUTO

ORA-00308: cannot open archived log '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_8_1200525254.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

ORA-00308: cannot open archived log '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_8_1200525254.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

SYS@orcl> recover database using backup controlfile;

ORA-00279: change 2325498 generated at 05/07/2025 23:17:09 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_8_1200525254.dbf

ORA-00280: change 2325498 for thread 1 is in sequence #8

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

/u01/app/oracle/oradata/ORCL/redo02.log

Log applied.

Media recovery complete.

-- 로그스위치가 일어난 경우

SYS@orcl> recover database using backup controlfile;

ORA-00279: change 2325498 generated at 05/07/2025 23:17:09 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_8_1200525254.dbf

ORA-00280: change 2325498 for thread 1 is in sequence #8

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

/u01/app/oracle/oradata/ORCL/redo02.log

ORA-00279: change 2328301 generated at 05/08/2025 00:14:33 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_9_1200525254.dbf

ORA-00280: change 2328301 for thread 1 is in sequence #9

ORA-00278: log file '/u01/app/oracle/oradata/ORCL/redo02.log' no longer needed for this recovery

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

-- 다시물어본다.

-- 다시물어본 경우의 시퀀스번호에 해당하는 번호의 로그 번호를

-- alert log에서 찾아 해당하는 리두로그 파일을 찾아 복구해야 한다.

-- 해당 로그번호가 되었는지는 랜덤으로 표기됨으로 alert log에서 찾아야 한다.

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

/u01/app/oracle/oradata/ORCL/redo03.log

Log applied.

Media recovery complete.

SYS@orcl> alter database open resetlogs;

Database altered.

과거 백업본은 사용할 수 없으며 일관성 있는 whole database backup 받아야 한다.

<<시나리오14>> Datafile, redo log file은 손상되지 않고 control file만 손상되었을 경우 복구(Trace)

```
SYS@orcl> shutdown immediate
SYS@orcl> startup
SYS@orcl> create table hr.emp4 tablespace users as select * from hr.employees;
```

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
		1	19 /u01/app/oracle/oradata/ORCL/redo01.log	10	ACTIVE
NO		2326371	2328870		
		2	20 /u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
NO		2328870	1.8447E+19		
		3	18 /u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
NO		2326368	2326371		

■ 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> shut abort
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size               905969664 bytes
Database Buffers           620756992 bytes
Redo Buffers                7876608 bytes
ORA-00205: error in identifying control file, check alert log for more info
```

-- Alert log file

```
ORA-205 signalled during: ALTER DATABASE MOUNT...
```

```
2025-05-08T00:41:31.464764+09:00
```

```
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_11163.trc:
```

```
ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
ORA-27037: unable to obtain file status
```

Linux-x86_64 Error: 2: No such file or directory

```
SYS@orcl> select status from v$instance;
```

STATUS

STARTED

```
SYS@orcl> show parameter control_files
```

NAME	TYPE	VALUE

control_files	string	/u01/app/oracle/oradata/ORCL/ /control01.ctl

```
SYS@orcl> shutdown abort
```

ORACLE instance shut down.

최근 백업 컨트롤 파일 찾아 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/noarch/control01.ctl /u01/app/oracle/oradata/ORCL/  
'backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
SYS@orcl> startup mount
```

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

Database mounted.

백업컨트롤파일 이용해서 trace 파일을 생성한(일자별로 생성) 후 컨트롤파일을 재생성해서 복구

```
SYS@orcl> alter database backup controlfile to trace as '/home/oracle/cre_con.sql';
```

Database altered.

```
SYS@orcl> ! ls -l /home/oracle/cre_con.sql
```

```
-rw-r--r--. 1 oracle dba 5828 May  8 00:44 /home/oracle/cre_con.sql
```

```
SYS@orcl> shutdown abort
```

ORACLE instance shut down.

컨트롤 파일 재생성의 이유(아래 값 변경시)

MAXLOGFILES 16

DB에서 생성가능한 리두로그 파일(그룹) 최대 갯수

MAXLOGMEMBERS 3

로그파일 그룹당 파일 최대 갯수

MAXDATAFILES 100

DB에서 생성가능한 데이터파일 최대 갯수 자동으로 증가 가능

MAXINSTANCES 8

인스턴스의 최대 갯수-RAC용

MAXLOGHISTORY 292

아카이브로그모드일시 최대 로그파일 기록 갯수

SYS@orcl> startup nomount

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

SYS@orcl> CREATE CONTROLFILE REUSE DATABASE "ORCL" NORESETLOGS NOARCHIVELOG

MAXLOGFILES 16

MAXLOGMEMBERS 3

MAXDATAFILES 100

MAXINSTANCES 8

MAXLOGHISTORY 292

LOGFILE

GROUP 1 '/u01/app/oracle/oradata/ORCL/redo01.log' SIZE 10M BLOCKSIZE 512,

GROUP 2 '/u01/app/oracle/oradata/ORCL/redo02.log' SIZE 10M BLOCKSIZE 512,

GROUP 3 '/u01/app/oracle/oradata/ORCL/redo03.log' SIZE 10M BLOCKSIZE 512

-- STANDBY LOGFILE

DATAFILE

'/u01/app/oracle/oradata/ORCL/system01.dbf',

'/u01/app/oracle/oradata/ORCL/sysaux01.dbf',

'/u01/app/oracle/oradata/ORCL/undotbs01.dbf',

'/u01/app/oracle/oradata/ORCL/users01.dbf'

CHARACTER SET AL32UTF8

;

Control file created.

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
MOUNTED
```

```
SYS@orcl> recover database;
```

```
Media recovery complete.
```

```
SYS@orcl> alter database open;
```

```
Database altered.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
		1	19 /u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
NO	2326371	2328870			
		2	20 /u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
NO	2328870	1.8447E+19			
		3	18 /u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
NO	2326368	2326371			

```
-- 백업본 그대로 사용 가능, 현재 리두로그파일 그대로 사용가능
```

```
-- 리두로그가 깨졌을때 RESETLOGS가 있는 부분을 사용
```

```
# 컨트롤파일 재생성 후에 꼭 체크
```

```
기존 tempfile 연결이 끊김으로 reues로 재사용하자.
```

```
SYS@orcl> select name from v$tempfile;
```

```
no rows selected
```

```
SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
-rw-r-----. 1 oracle dba 33562624 May 8 00:59 /u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
SYS@orcl> ALTER TABLESPACE TEMP ADD TEMPFILE '/u01/app/oracle/oradata/ORCL/temp01.dbf' REUSE;
```

```
Tablespace altered.
```

```
SYS@orcl> select name from v$tempfile;
```

```
NAME
```

```
-----
```

/u01/app/oracle/oradata/ORCL/temp01.dbf

- 원래대로 돌리기

SYS@orcl> shutdown abort

ORACLE instance shut down.

SYS@orcl> ! cp -v backup/noarch/*.{ctl,dbf,log} /u01/app/oracle/oradata/ORCL/

'backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'

'backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'

'backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'

'backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'

'backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

'backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'

'backup/noarch/redo05.log' -> '/u01/app/oracle/oradata/ORCL/redo05.log'

'backup/noarch/redo06.log' -> '/u01/app/oracle/oradata/ORCL/redo06.log'

'backup/noarch/redo07.log' -> '/u01/app/oracle/oradata/ORCL/redo07.log'

<<시나리오 15>> datafile, redo log file 은 손상되지 않고 control file 만 손상되었을 경우 복구(정상종료)

-- 로그스위치 많이 발생해서 넘어갔어도 정상적인 종료가 되었다가 재시작시 컨트롤 파일 문제가 발생한 경우

```
SYS@orcl> create table hr.emp3 tablespace users as select * from hr.employees;
Table created.
```

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
		1	22 /u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
NO		2532029	1.8447E+19		
		2	20 /u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE
NO		2328870	2528932		
		3	21 /u01/app/oracle/oradata/ORCL/redo03.log	10	ACTIVE
NO		2528932	2532029		

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

■ 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

ORA-00205: error in identifying control file, check alert log for more info

```
# 최근 백업 컨트롤파일 백업본을 찾아 restore
```

```
SYS@orcl> ! cp -v backup/noarch/control01.ctl /u01/app/oracle/oradata/ORCL/  
'backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
SYS@orcl> alter database mount;  
Database altered.
```

```
SYS@orcl> alter database backup controlfile to trace as '/home/oracle/backup/cre_cont.sql';  
Database altered.
```

```
SYS@orcl> shutdown abort
```

```
SYS@orcl> startup nomount
```

```
SYS@orcl> CREATE CONTROLFILE REUSE DATABASE "ORCL" NORESETLOGS NOARCHIVELOG  
MAXLOGFILES 16  
MAXLOGMEMBERS 3  
MAXDATAFILES 100  
MAXINSTANCES 8  
MAXLOGHISTORY 292
```

```
LOGFILE
```

```
GROUP 1 '/u01/app/oracle/oradata/ORCL/redo01.log' SIZE 10M BLOCKSIZE 512,  
GROUP 2 '/u01/app/oracle/oradata/ORCL/redo02.log' SIZE 10M BLOCKSIZE 512,  
GROUP 3 '/u01/app/oracle/oradata/ORCL/redo03.log' SIZE 10M BLOCKSIZE 512  
-- STANDBY LOGFILE
```

```
DATAFILE
```

```
'/u01/app/oracle/oradata/ORCL/system01.dbf',  
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf',  
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf',  
'/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
CHARACTER SET AL32UTF8
```

```
;
```

```
Control file created.
```

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
MOUNTED
```

```
SYS@orcl> alter database open;
```

```
Database altered.
```

```
SYS@orcl> ALTER TABLESPACE TEMP ADD TEMPFILE '/u01/app/oracle/oradata/ORCL/temp01.dbf' REUSE;
```


<<시나리오 16>> datafile, redo log file 은 손상되지 않고 control file 만 손상되었을 경우
복구(비정상종료)

SYS@orcl> create table hr.emp4 tablespace users as select * from hr.employees;

■ 장애 발생

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl

SYS@orcl> alter system switch logfile;

System altered.

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#			
	1	22	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
NO	2532029	2532205			
	2	23	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
NO	2532205	1.8447E+19			
	3	21	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
NO	2528932	2532029			

<<hr>>

HR@orcl> select count(*) from employees;

```

COUNT(*)
-----
107

```

HR@orcl> conn / as sysdba

Connected.

SYS@orcl> select * from v\$database; --> 문제없다면 Alert log 확인

select * from v\$database

*

ERROR at line 1:

ORA-00210: cannot open the specified control file

ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'

ORA-27041: unable to open file

Linux-x86_64 Error: 2: No such file or directory

Additional information: 3

```
[oracle@oel7v9r1 trace]$ vi alert_orcl.log
```

```
-- Alert log
```

```
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz01_18408.trc:
```

```
ORA-00210: cannot open the specified control file
```

```
ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
ORA-27041: unable to open file
```

```
Linux-x86_64 Error: 2: No such file or directory
```

Additional information: 3

```
SYS@orcl> shutdown abort
```

```
ORACLE instance shut down.
```

```
# 최근 백업 컨트롤파일 백업본을 찾아 restore
```

```
SYS@orcl> shut abort
```

```
ORACLE instance shut down.
```

```
SYS@orcl> ! cp -v /home/oracle/backup/noarch/control01.ctl /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

```
Fixed Size 8895728 bytes
```

```
Variable Size 754974720 bytes
```

```
Database Buffers 369098752 bytes
```

```
Redo Buffers 7880704 bytes
```

```
Database mounted.
```

```
ORA-01122: database file 1 failed verification check
```

```
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-01207: file is more recent than control file - old control file
```

```
-- 백업 컨트롤파일과 현재 데이터파일의 SCN 번호가 달라 컨트롤파일이 오래됨
```

```
SYS@orcl> alter database backup controlfile to trace as '/home/oracle/backup/cons.sql';
```

```
SYS@orcl> shutdown abort
```

```
ORACLE instance shut down.
```

```
SYS@orcl> STARTUP NOMOUNT
```

```
CREATE CONTROLFILE REUSE DATABASE "ORCL" NORESETLOGS NOARCHIVELOG
```

```
MAXLOGFILES 16
```

```
MAXLOGMEMBERS 3
```

```
MAXDATAFILES 100
```

```
MAXINSTANCES 8
```

MAXLOGHISTORY 292

LOGFILE

GROUP 5 '/u01/app/oracle/oradata/ORCL/redo05.log' SIZE 50M BLOCKSIZE 512,

GROUP 6 '/u01/app/oracle/oradata/ORCL/redo06.log' SIZE 50M BLOCKSIZE 512,

GROUP 7 '/u01/app/oracle/oradata/ORCL/redo07.log' SIZE 50M BLOCKSIZE 512

DATAFILE

'/u01/app/oracle/oradata/ORCL/system01.dbf',

'/u01/app/oracle/oradata/ORCL/undotbs01.dbf',

'/u01/app/oracle/oradata/ORCL/sysaux01.dbf',

'/u01/app/oracle/oradata/ORCL/users01.dbf'

CHARACTER SET AL32UTF8

;

Control file created.

control file 손상 후에 데이터베이스가 비정상적인 종료 발생했기 때문에 꼭 recover database 명령어 수행해야 한다.

SYS@orcl> recover database;

Media recovery complete.

SYS@orcl> alter database open;

Database altered.

SYS@orcl> ALTER TABLESPACE TEMP ADD TEMPFILE '/u01/app/oracle/oradata/ORCL/temp01.dbf' REUSE;

Tablespace altered.

<<시나리오17>> datafile, control file 손상되었을 경우 복구(데이터베이스 정상 종료, 리두정보 O)

```
SYS@orcl> create table hr.emp4 tablespace users as select * from hr.employees;
```

Commit complete.

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

■ 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,dbf
```

```
/u01/app/oracle/oradata/ORCL/control01.ctl
```

```
/u01/app/oracle/oradata/ORCL/control02.ctl
```

```
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
```

```
/u01/app/oracle/oradata/ORCL/system01.dbf
```

```
/u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.ctl: No such file or directory
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.dbf: No such file or directory
```

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

ORA-00205: error in identifying control file, check alert log for more info

-- Alert log file

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_20358.trc:

ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/  
initorcl.ora redo01.log redo02.log redo03.log
```

문제가 발생한 control, data file 백업 찾아서 restore

```
SYS@orcl> shutdown abort
```

ORACLE instance shut down.

```
SYS@orcl> ! cp -v /home/oracle/backup/noarch/*.ctl,dbf /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'  
'/home/oracle/backup/noarch/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'  
'/home/oracle/backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'  
'/home/oracle/backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'  
'/home/oracle/backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'  
'/home/oracle/backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'  
'/home/oracle/backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SYS@orcl> startup mount
```

```
SYS@orcl> recover database using backup controlfile;
```

ORA-00279: change 2326564 generated at 05/08/2025 01:00:31 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_19_1200525254.dbf

ORA-00280: change 2326564 for thread 1 is in sequence #19

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

auto

ORA-00308: cannot open archived log '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_19_1200525254.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

ORA-00308: cannot open archived log '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_19_1200525254.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		
1	19	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
NO	2326371	1.8447E+19		

	2	17 /u01/app/oracle/oradata/ORCL/redo02.log	10 INACTIVE
NO	2326365	2326368	
	3	18 /u01/app/oracle/oradata/ORCL/redo03.log	10 INACTIVE
NO	2326368	2326371	

SYS@orcl> recover database using backup controlfile;

ORA-00279: change 2326564 generated at 05/08/2025 01:00:31 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_19_1200525254.dbf

ORA-00280: change 2326564 for thread 1 is in sequence #19

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

/u01/app/oracle/oradata/ORCL/redo01.log

Log applied.

Media recovery complete.

SYS@orcl> alter database open resetlogs;

Database altered.

--백업 합시다.